Implementation and Evaluation of Wind Profile Estimation Algorithms

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Project Overview

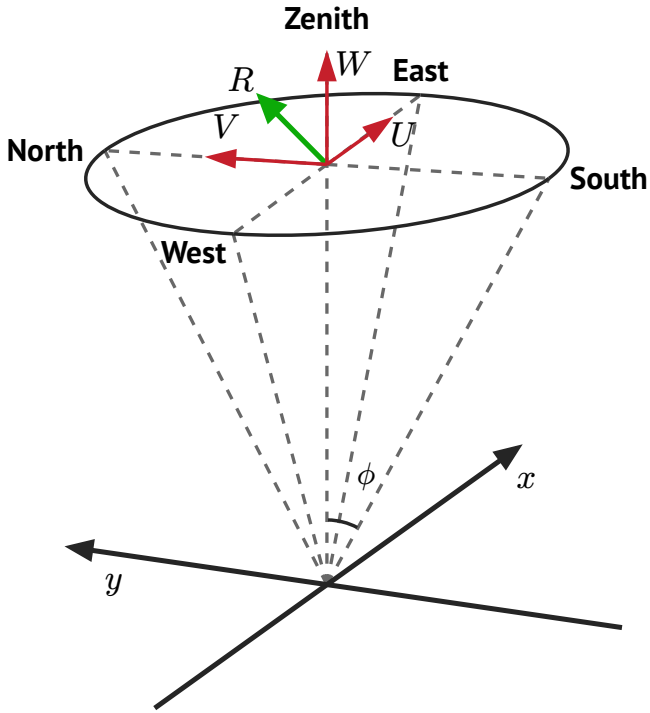
Project Guide: Dr Binu Paul
Project Phase: Phase 1
Research Area: Radar Signal Processing
Focus: Evaluation and comparison of MCCF, IME and hybrid method

Presentation Outline

1. Introduction
   - Introduction to Wind Profiler Radar
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Introduction to Wind Profiler Radar

- Wind profiler radars use electromagnetic waves to estimate atmospheric wind profiles remotely.
- The Doppler shift of returned atmospheric echoes provides information about radial wind velocity.
- Doppler power spectra are used to estimate important parameters:
  - M_0 – Signal power
  - M_1 – Mean Doppler velocity
  - M_2 – Spectral width
- This project focuses on the **205 MHz ST radar at CUSAT**



Existing Problem

- Wind velocity estimation using the 205 MHz ST radar at CUSAT requires accurate identification of peaks in Doppler spectra.
- **Major challenges:**
  - Low signal-to-noise ratio (SNR)
  - Ground clutter and unwanted interference
  - Multiple spectral peaks and ambiguous peak selection
- Incorrect peak detection leads to errors in Doppler velocity estimation and discontinuities in wind profiles.
- **Objective:** Compare MCCF, IME, and hybrid methods to study their accuracy under different radar signal conditions.

Project Overview

Research Focus

Reliable Doppler spectral peak and moment estimation for the 205 MHz ST radar at CUSAT.

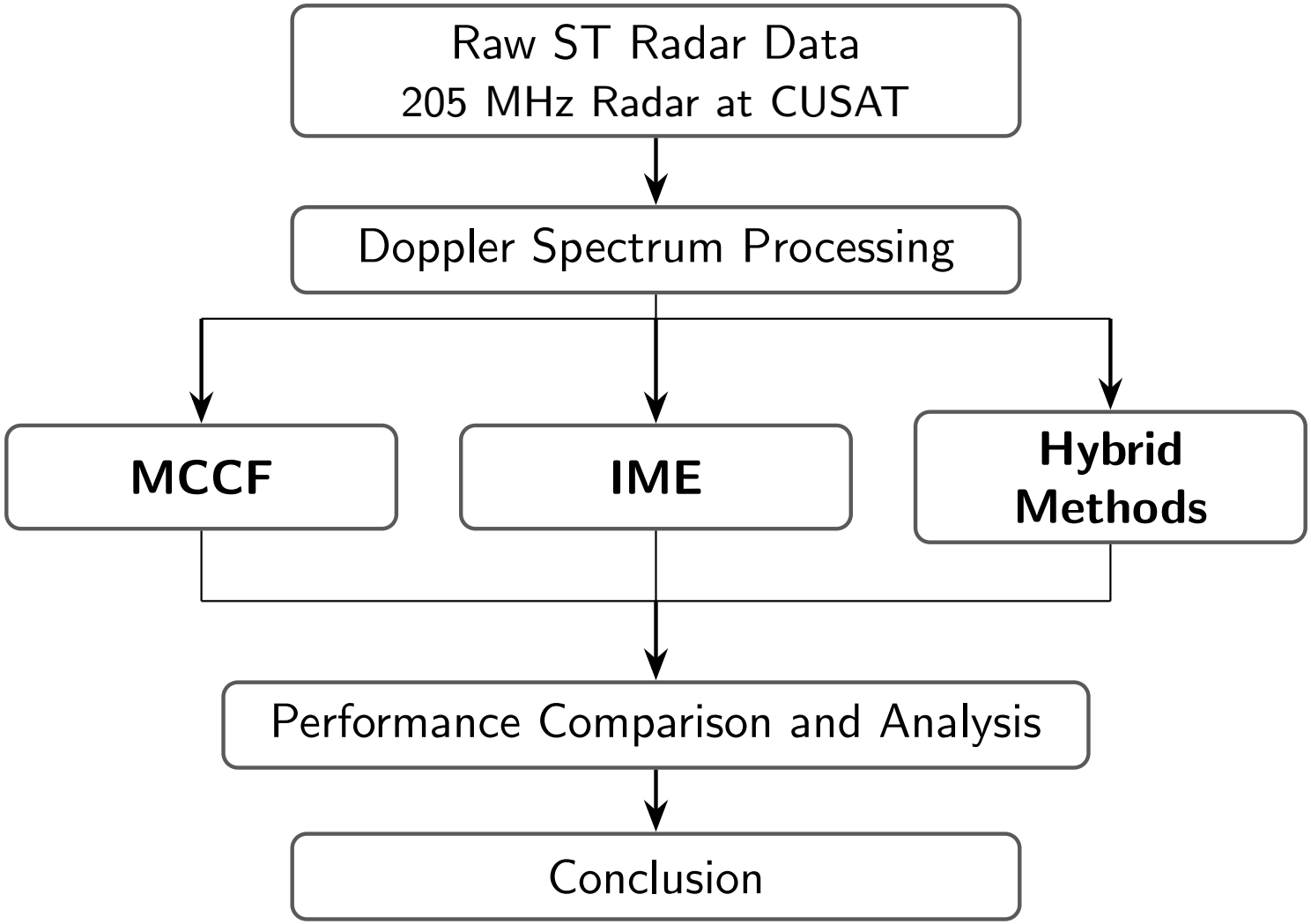
Methods Investigated

- **MCCF:** Mutual Convolution Cost Function
- **IME:** Improved Moments Estimation
- **Hybrid Methods:** Combining mutual convolution with IME processing (Hybrid 1 and Hybrid 2)

Main Objective

- Evaluate the methods on radar spectra.
- Evaluate performance using synthetic data with known parameters.

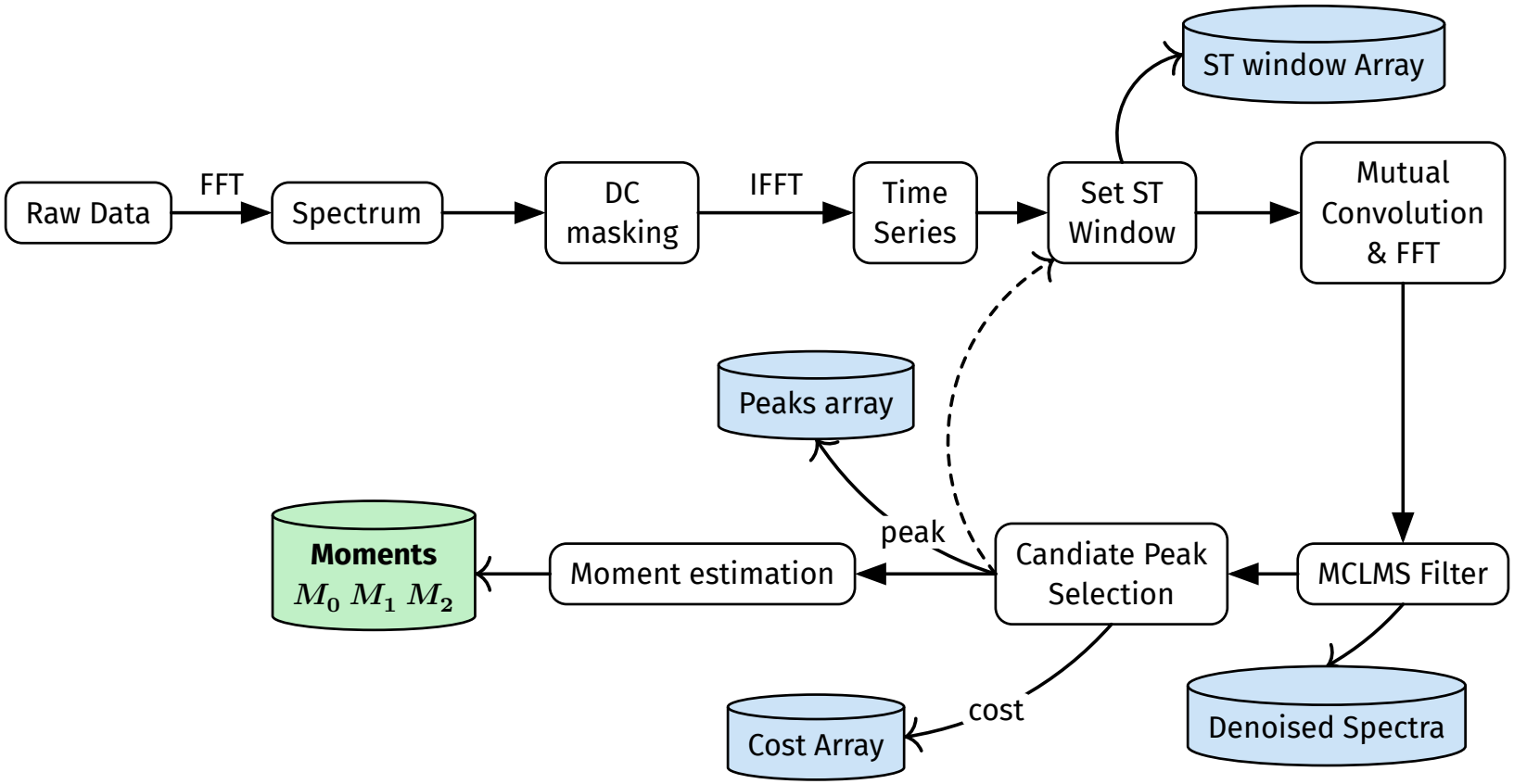
Project Workflow



MCCF: Working Principle

- A Doppler spectral peak estimation method with noise suppression.
- Processing Stages
  - **Preprocessing:** FFT and DC-component removal.
  - **Noise Suppression:** Mutual convolution and MCLMS adaptive filtering.
  - **Peak Selection:** Candidate peaks are evaluated using a velocity-continuity and magnitude-based cost function.
  - **Moment Estimation:** Estimate the Doppler spectral moments from the selected peak.
- Improve spectral quality and select a suitable peak instead of relying only on the strongest spectral component.

MCCF: Algorithm Flow



MCCF: Important steps

- **Setting Space-Time window**: Limits peak search window to a selected Doppler range (space). (No time window is used due to the large time difference between observations)
- **Mutual Convolution**: Divides the single spectrum into slices of half length and convolves them together to generate multiple intermediate combined results. Pick the result with the highest peak
- **Modified Complex Least Mean Square Filter**: Isolate the stable, coherent features and remove random noise and ground clutter that fluctuate unpredictably across different delay steps

MCCF: Results and Observations

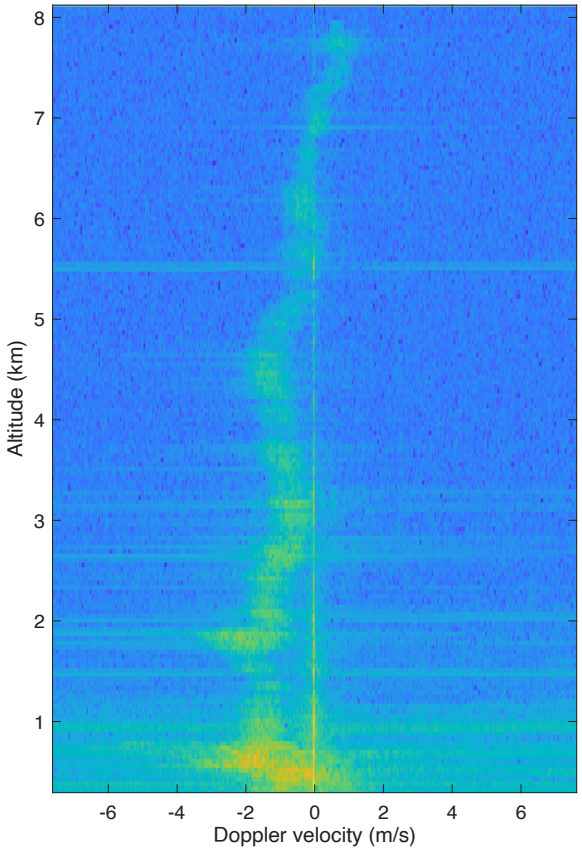


Figure 1: Original Spectrum

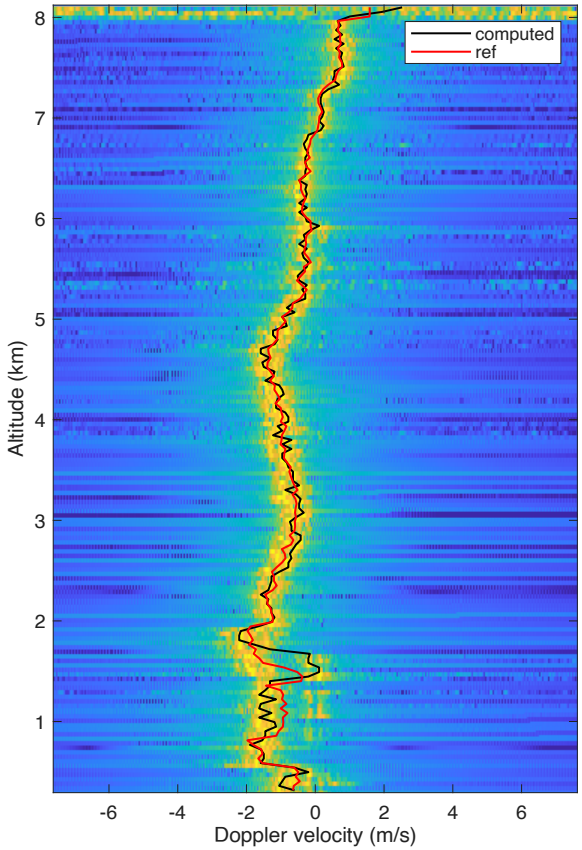


Figure 2: Estimated Moments

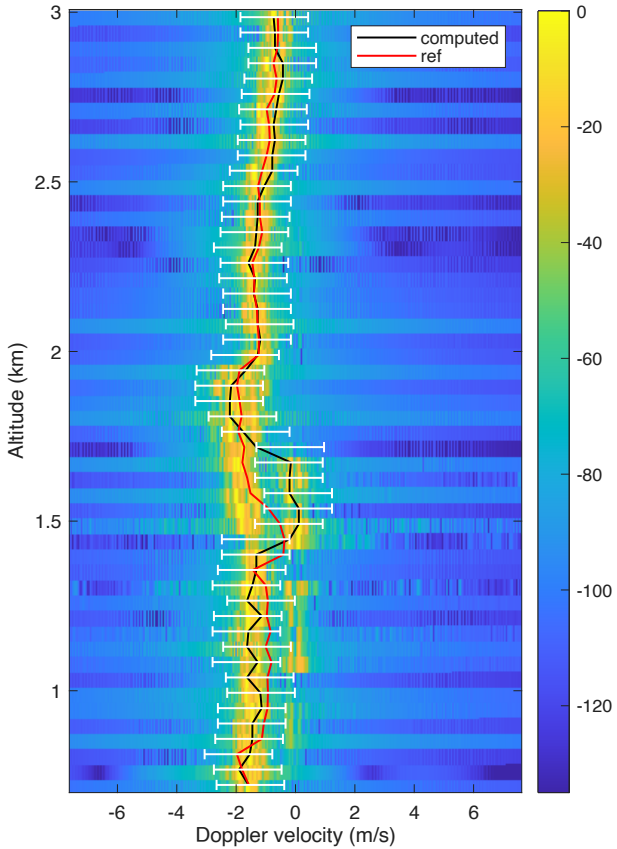


Figure 3: ST window

IME: Working Principle

A Doppler spectral estimation method designed to identify reliable atmospheric peaks under noisy and low-SNR conditions.

Main Processing Stages

- Spectral Preprocessing
  - Remove DC components and ground clutter.
  - Apply 5-point smoothing and estimate the noise level.
- Adaptive Doppler Window
  - Restrict peak searching to a selected Doppler range.
  - Identify prospective peaks using their SNR.
- Profile Tracing
  - Connect candidate peaks across adjacent range bins.
  - Select the profile using velocity continuity and relative SNR.

(a): Preprocessing

- **DC and Ground Clutter Removal**
  - Suppress the zero-frequency component by interpolating neighbouring frequency bins.
- **Smoothing and Filtering**
  - Apply a five-point moving-average filter to reduce transient interference.
- **Noise Estimation and Reduction**
  - Estimate noise power using the Hildebrand–Sekhon method.
  - Subtract the estimated noise and clip negative values to zero.

(b): Adaptive Doppler Window

Adaptive Window Selection

- Start from the lowest range bin.
- Use its maximum peak as the initial mean Doppler velocity.
- Centre the next Doppler window around the candidate closest to the previous mean velocity.

Prospective Peak Selection

- Detect peaks within the adaptive window
- Retain peaks with SNR > 0 dB (upto 3 Peaks).

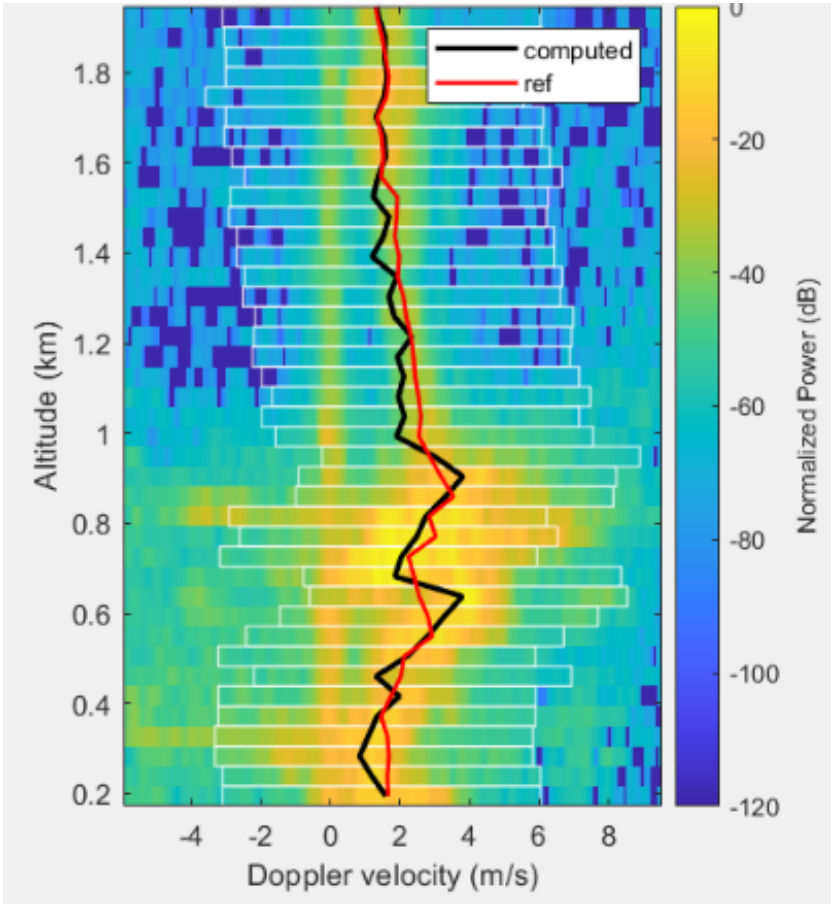


Figure 4: Adaptive Doppler window

(c): Grouping and Cost Allocation

1. Grouping and Link Construction
2. Cost Function: The link cost combines velocity continuity and normalised SNR:

$$C = w_1 T + w_2 P$$

3. Final Profile Selection
   - Select the trace with the maximum total cost.
   - Use its final velocity to initialise the next group.

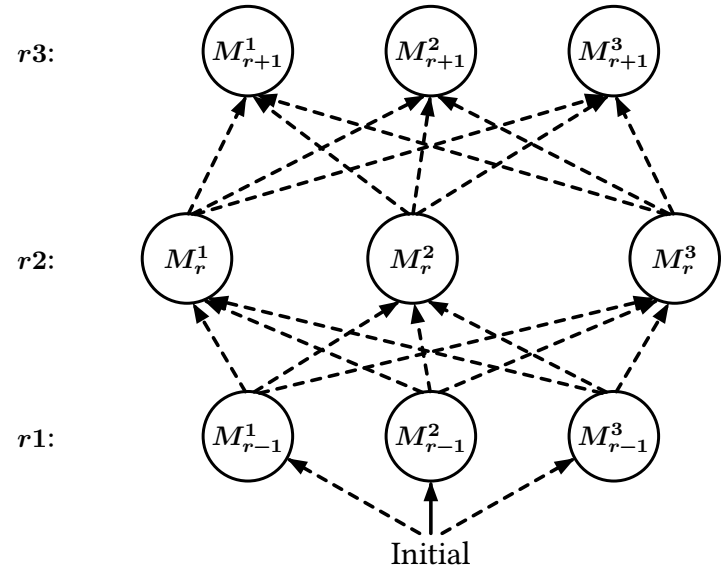


Figure 5: Grouping and profile tracing

IME: Results

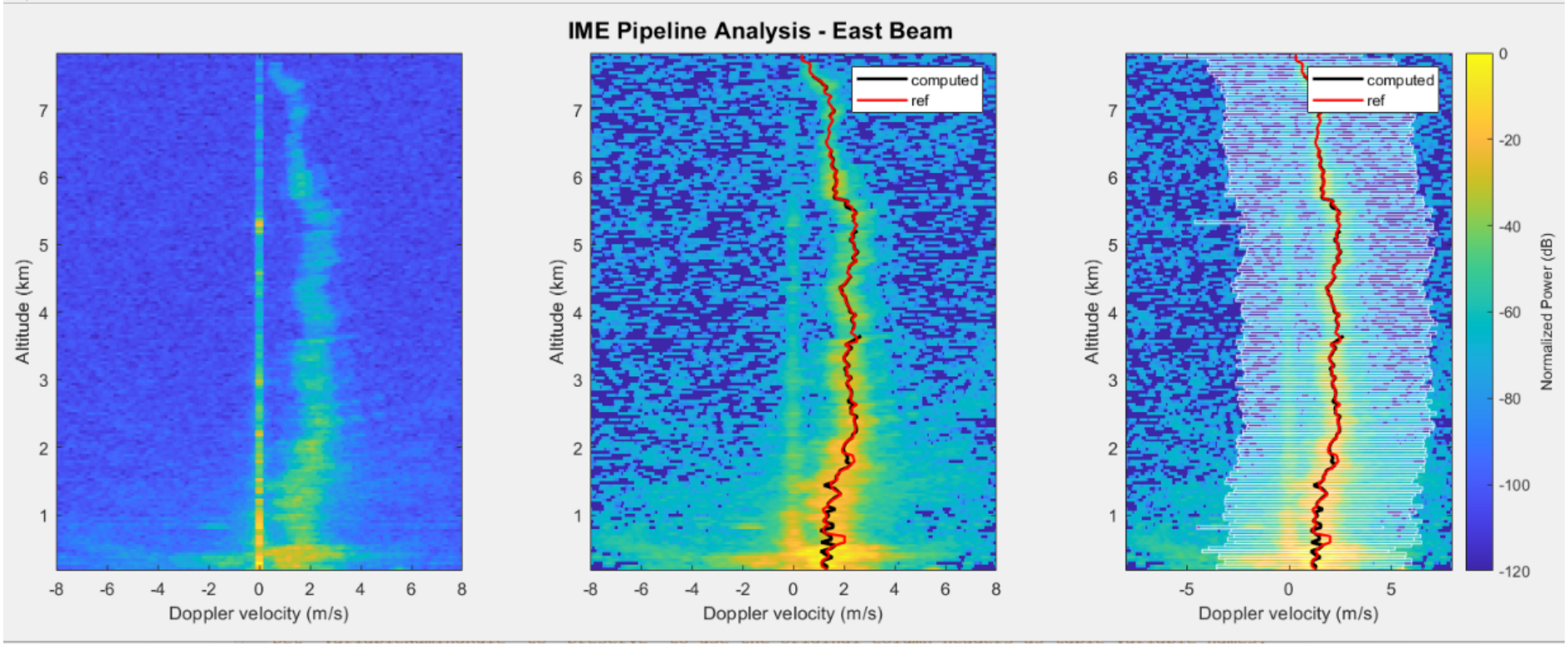


Figure 6: a) Original b) After applying IME c) Adaptive window

Hybrid Methods

Mutual convolution (MC) in MCCF provides noise suppression, while IME provides adaptive peak selection and profile tracing. The hybrid methods combine these processing stages.

Hybrid 1

MC + IME Denoising + IME Peak Selection

- Spectrum processing using MC
- IME denoising
- IME peak selection

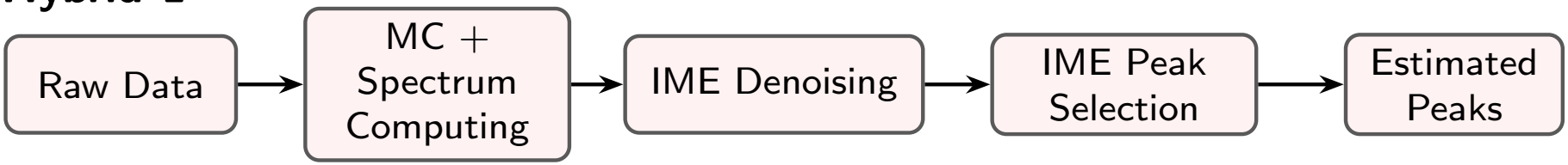
Hybrid 2

MC + IME Peak Selection

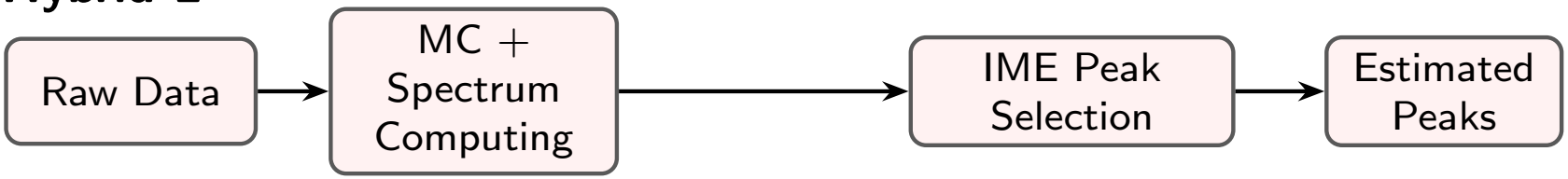
- Spectrum processing using MC
- Direct IME peak selection

Hybrid Methods: Flowchart

Hybrid 1

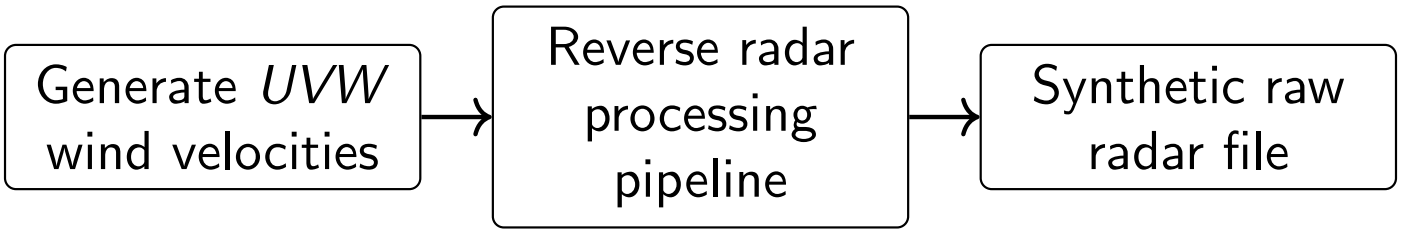


Hybrid 2



Synthetic Radar Data Generation

Generated from a predefined *UVW* wind-velocity array by reversing the conventional radar signal-processing pipeline to produce a raw data file.



- Provides known ground-truth wind velocities.
- Enables direct comparison of algorithm outputs against the predefined velocities.
- Allows controlled testing of algorithms without relying exclusively on newly acquired radar observations.

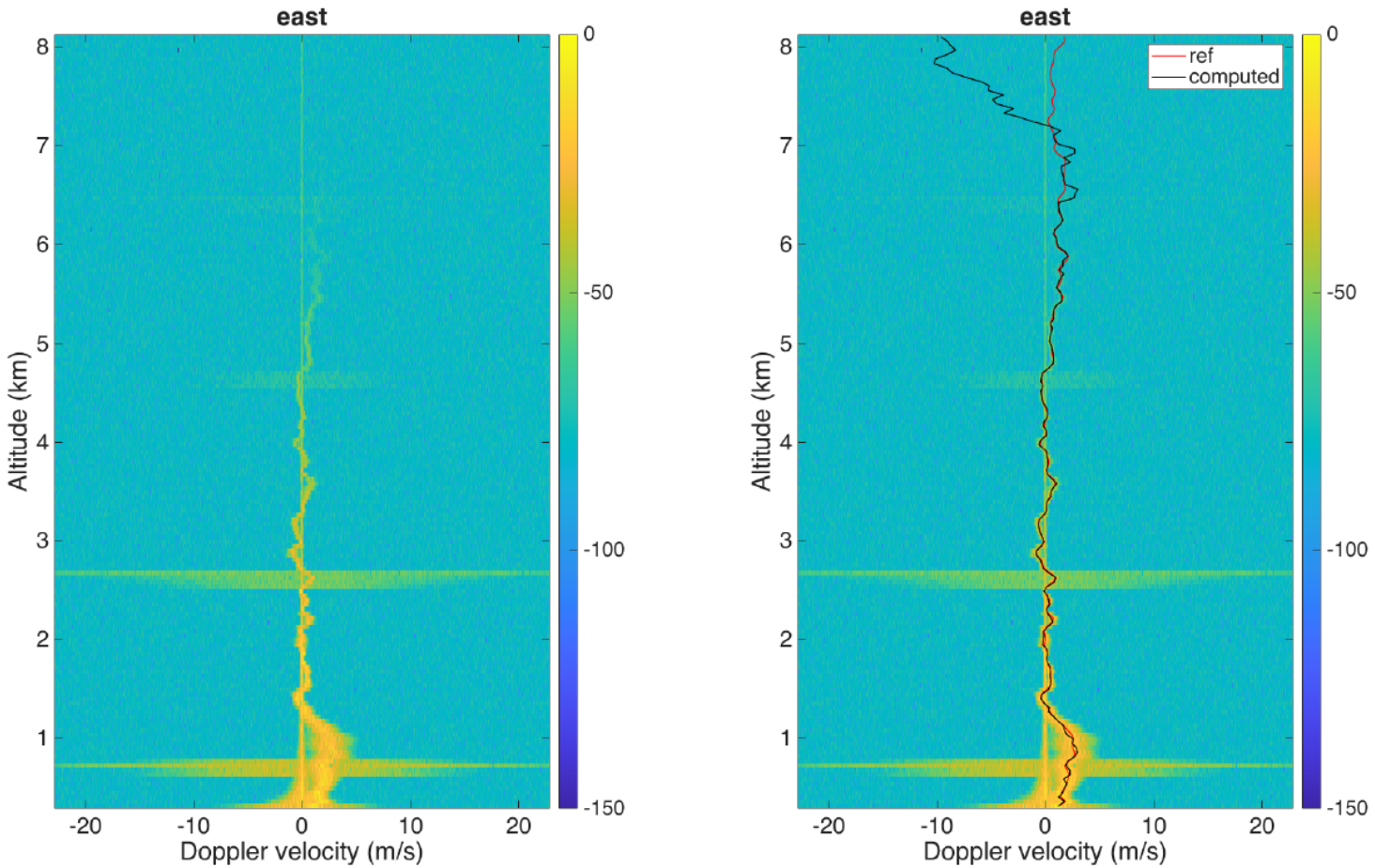


Figure 7: Synthesised raw data and corresponding velocity estimation using MCCF

Evaluvation with Synthetic data

Synthesised 500 raw data samples and applied it to 5 algorithms.

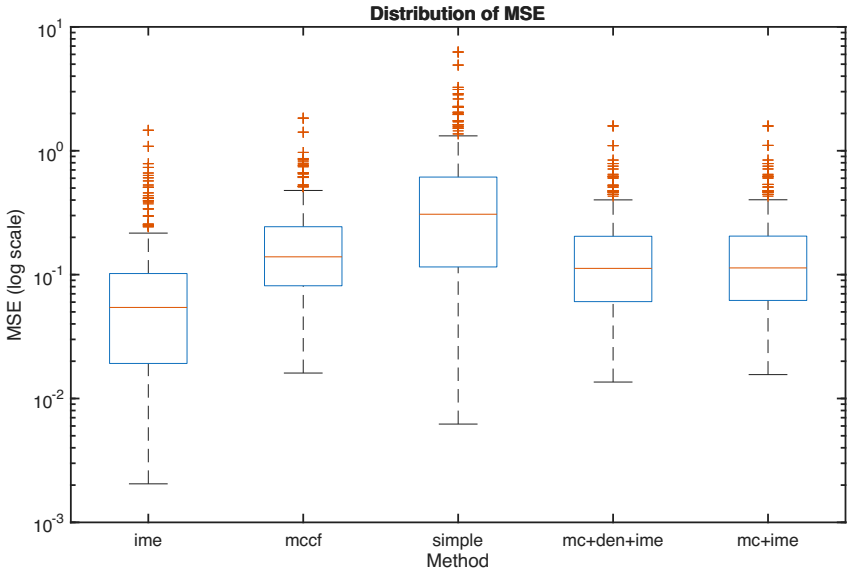


Figure 8: Mean Squared Error (MSE)

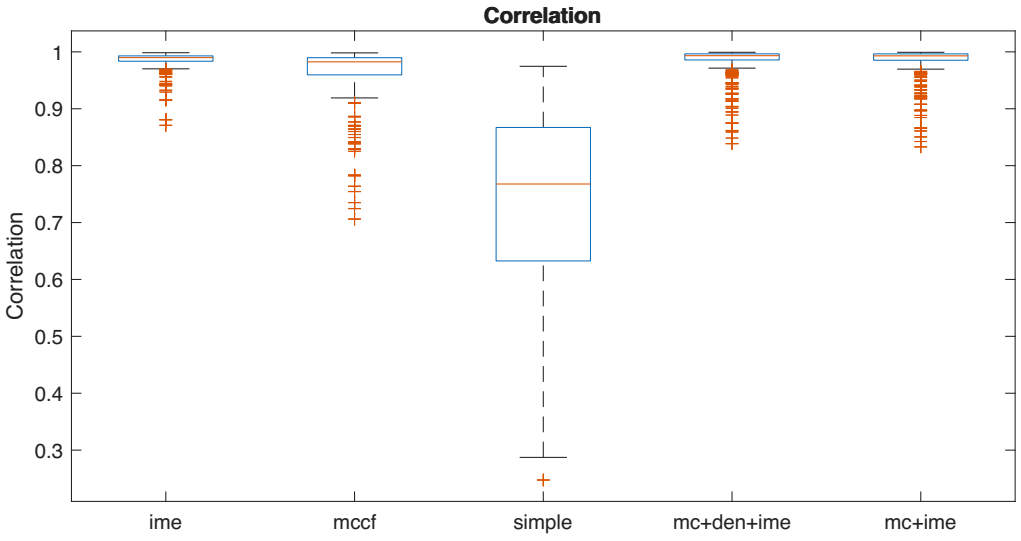


Figure 9: Correlation with ref data

Evaluvation with Synthetic data

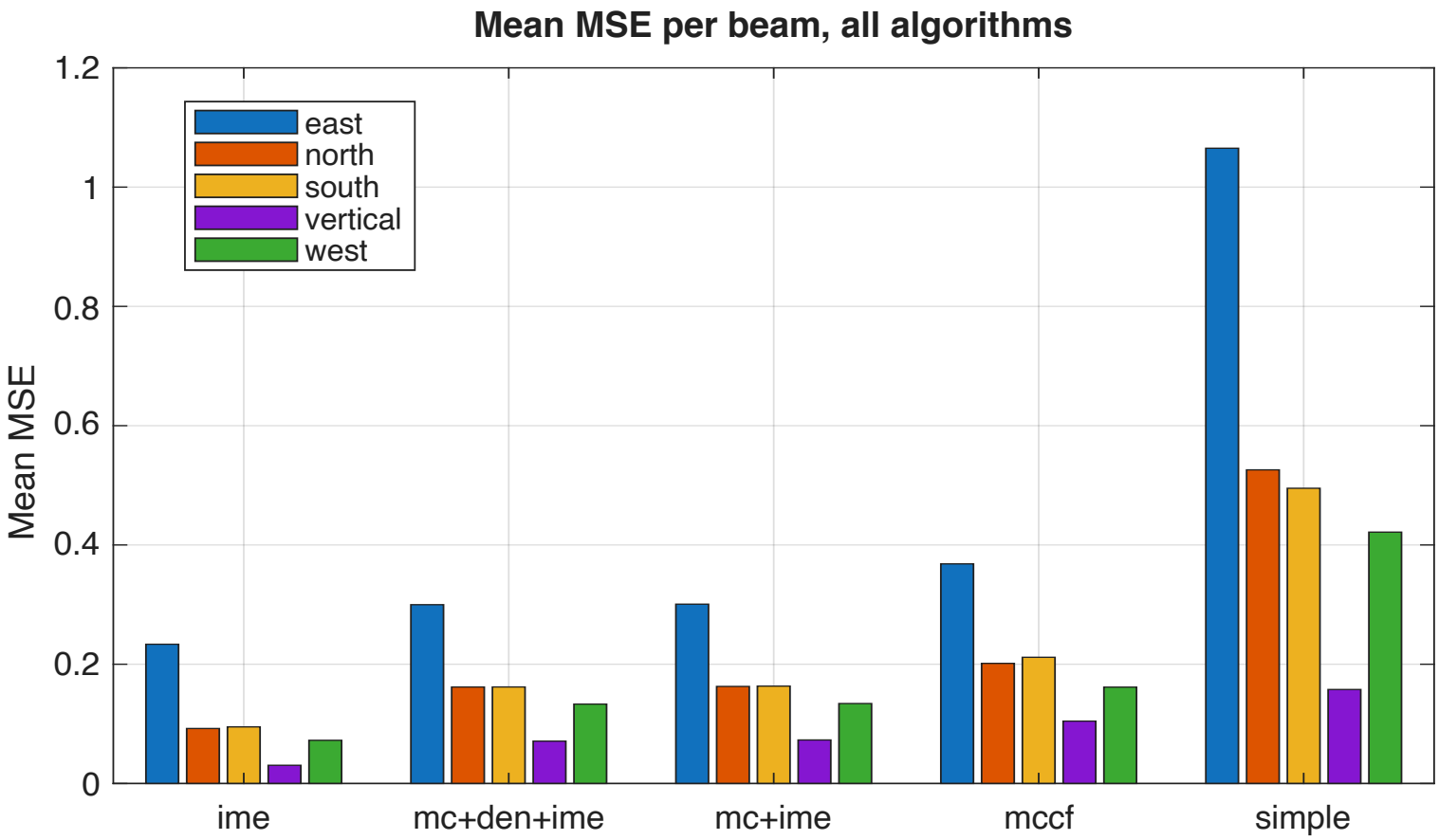


Figure 10: MSE in all beams

Evaluvation with Radiosonde

- Projection of radiosonde observation along with MCCF's estimation.
- In the winter season, the atmosphere is stable hence, the result closely matches

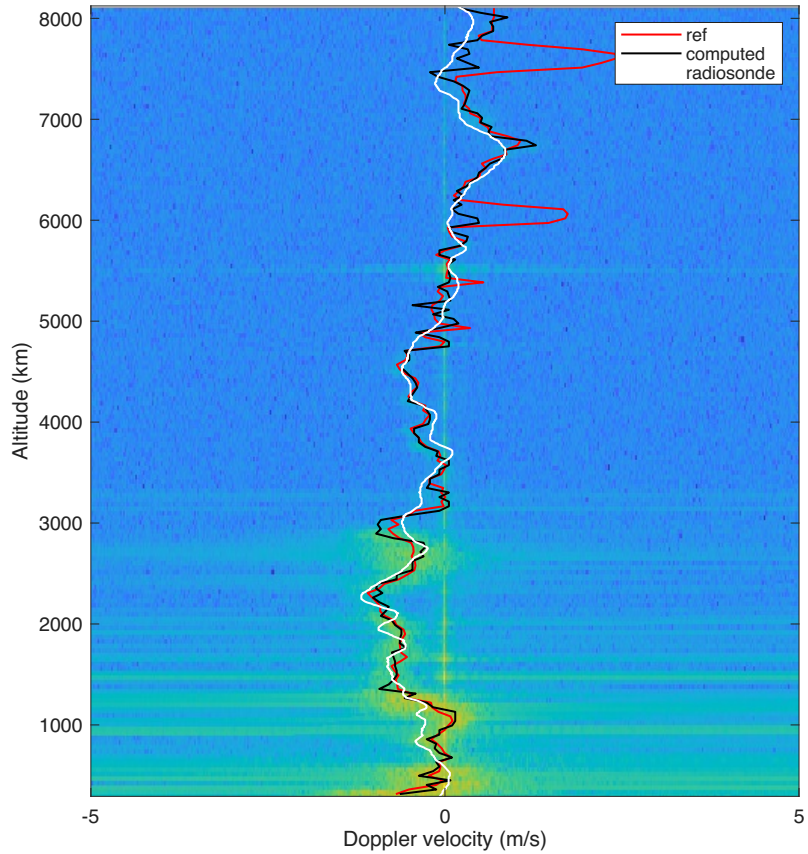


Figure 11: Observation on 20 Jan 2026, 14:20 IST

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Evaluvation with Data Pattern's output

Used 226 radar observations taken on 20 January 2026.

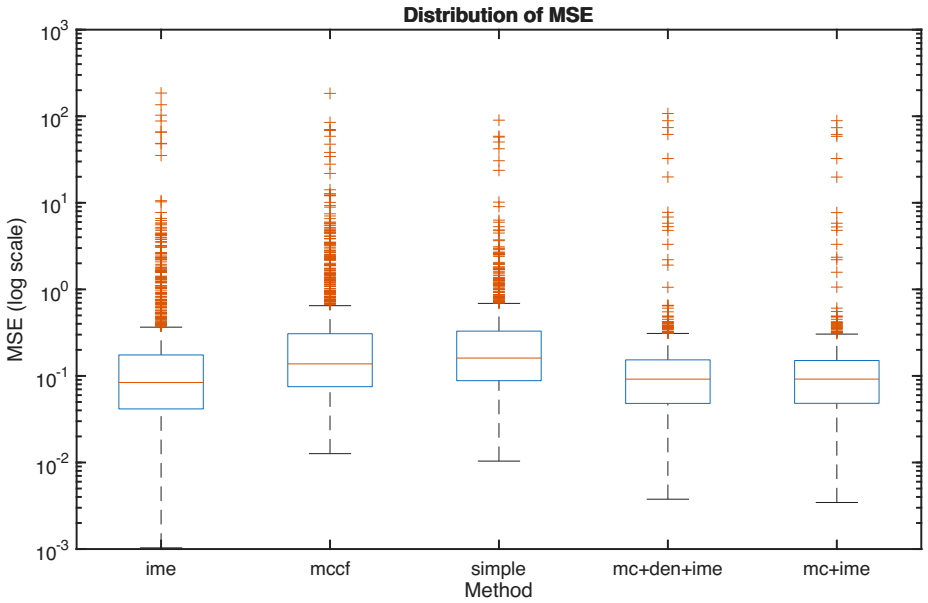


Figure 12: MSE

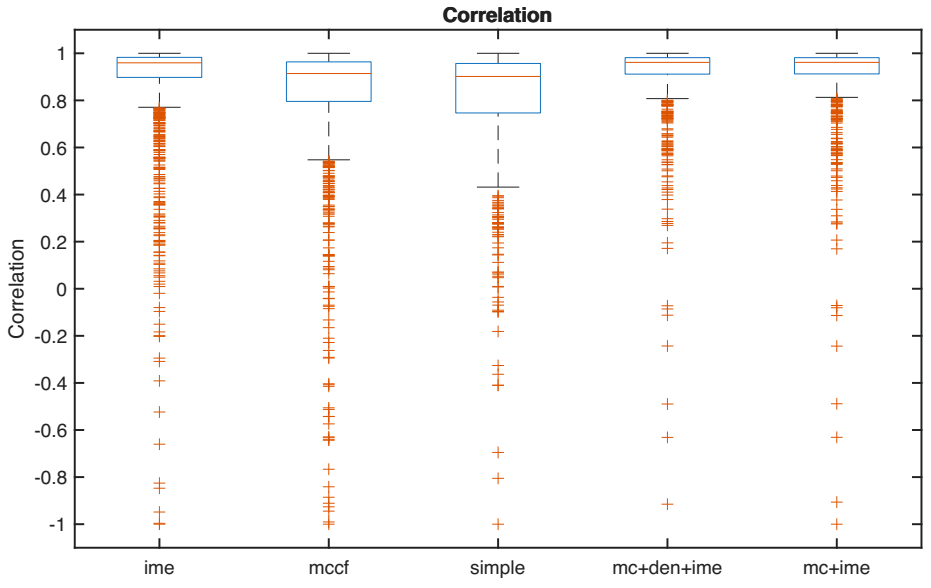


Figure 13: Correlation with reference data

Evaluvation with Data Pattern's output

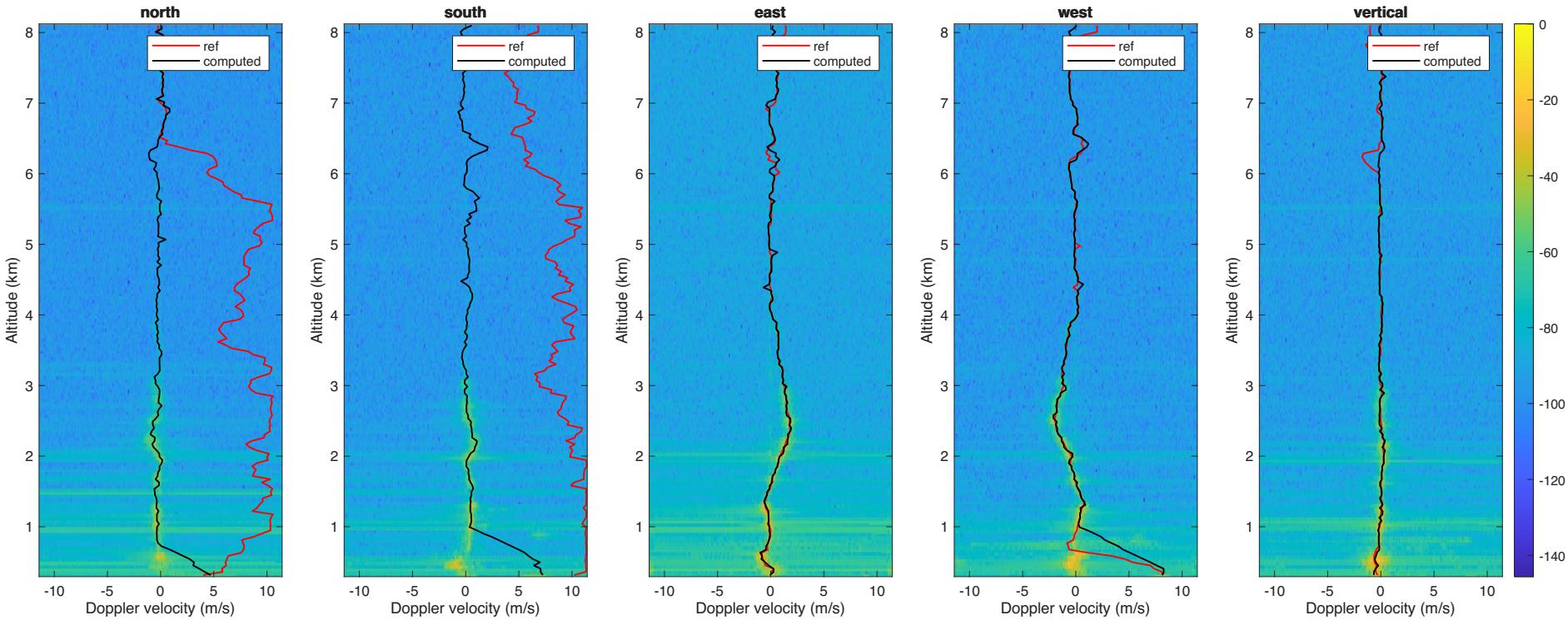


Figure 14: Case where Data Pattern's algorithm fell short, moments computed by MCCF

Conclusion

- Studied MCCF and IME for Doppler spectral peak and moment estimation using ST radar data.
- Examined two hybrid approaches combining mutual convolution with IME peak-selection and profile-tracing stages.
- The methods address challenges such as low SNR and unwanted spectral components.
- Methods were evaluated with various sources.

Future Scope

- Wavelet-based denoising: Improve weak Doppler peak detection under low-SNR conditions.
- ML-Based Peak Detection: Improve atmospheric echo identification in multi-peak spectra.
- Making synthesised radar data more realistic by adding more features
- Distinguish rain echoes from clear-air atmospheric echoes in Doppler spectra to improve the reliability of wind-velocity estimation
- Can be used to fill gaps between observations in the lower atmosphere while the experiment is being conducted at higher altitudes
- Use synthetic data to train reinforcement learning-based algorithms for estimating wind profiles.

References I



Z. Tan, G. Yang, S. Wang, L. Wang, Y. Fan, R. Wang, and Y. Luo, "Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method," *IEEE Sensors Journal*, vol. 24, no. 15, pp. 24044–24053, Aug. 2024. doi: 10.1109/JSEN.2024.3405665.



Y. Li, R. Wang, Y. Luo, H. Li, and G. Yang, "Improved Moments Estimation for Ground-Based K-Band Doppler Radar Using Cost Function Method," *IEEE Access*, vol. 10, pp. 32049–32059, 2022. doi: 10.1109/ACCESS.2022.3159690.

THANK YOU